MAE 1001: Python “Power Chords”

Introduction to Mechanical and Aerospace Engineering

Course website: https://gwu-mae1001.github.io/fall2025/

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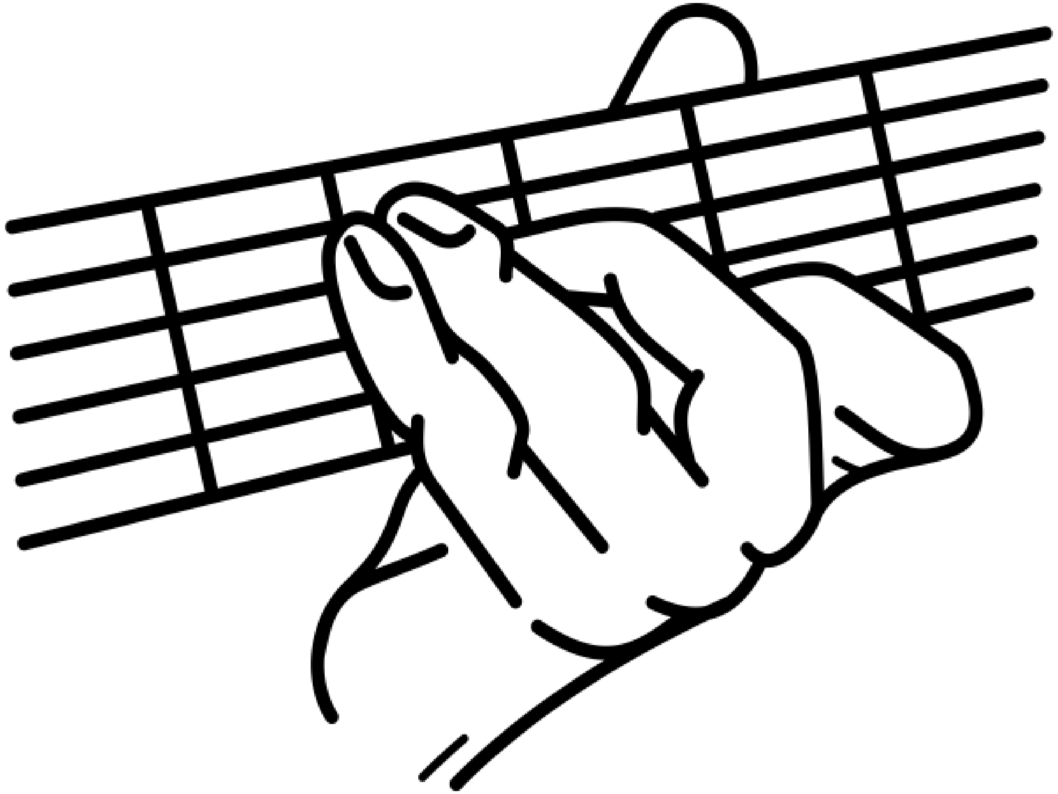
Fall 2025

Photo: Kartik Bulusu

"Study hard what interests you the most in the most undisciplined, irreverent and original manner possible."

— Richard Feynmann



What is a Power Chord?

- A power chord consists of just two notes: the root and the fifth.
- Commonly used in rock music for its strong, clear sound.

Representing Power Chords in Python

- Musical intervals can be modeled with arrays/lists, making it easy to calculate the root and fifth for any note.

Try to code using Thonny IDE that you all installed successfully!

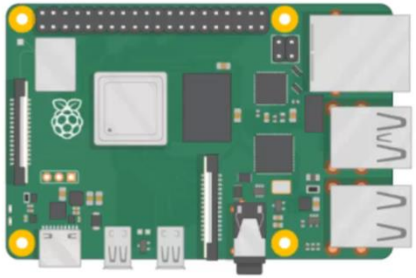
```
import numpy as np
import sounddevice as sd

def play_power_chord(root_freq, duration=1.0, fs=44100):
    fifth_freq = root_freq * 2**(7/12)
    t = np.linspace(0, duration, int(fs*duration), endpoint=False)
    tone = np.sin(2*np.pi*root_freq*t) + np.sin(2*np.pi*fifth_freq*t)
    tone = tone / np.max(np.abs(tone))  # Normalize
    sd.play(tone, fs)
    sd.wait()

play_power_chord(82.41)
play_power_chord(110)
play_power_chord(146.83)
play_power_chord(196)
play_power_chord(246.94)
play_power_chord(329.63)
```

How did it sound?

Sources:
Thonny (IDE): https://en.wikipedia.org/wiki/Thonny
Thonny: The Beginner-Friendly Python Editor: https://realpython.com/python-thonny/

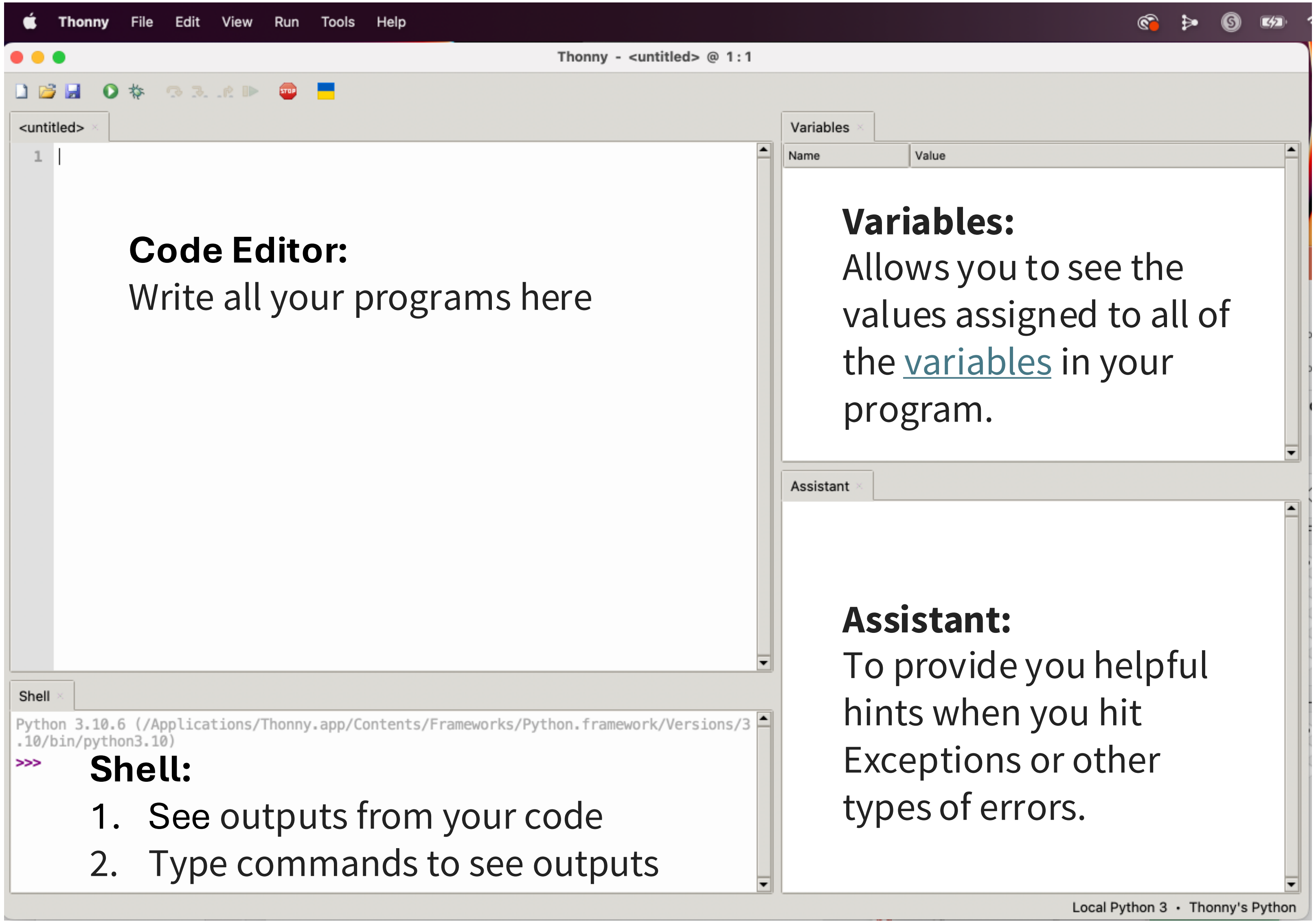
Thonny-editor or integrated development environment (IDE)

Th



GET STARTED WITH RASPBERRY PI

Source: https://www.raspberrypi.org/help/

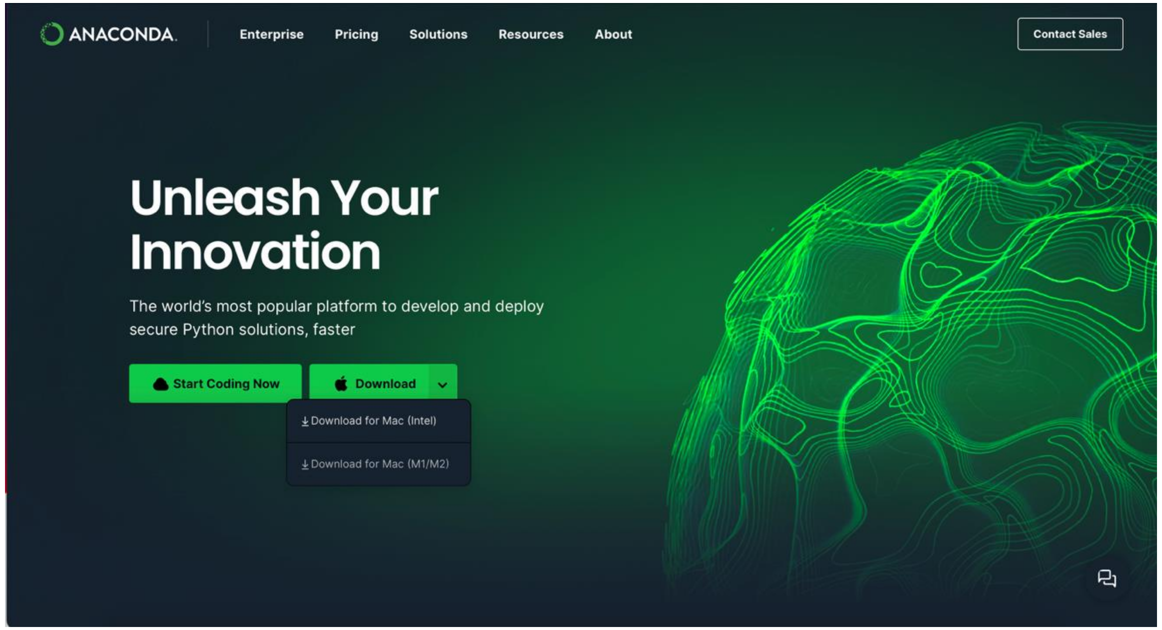


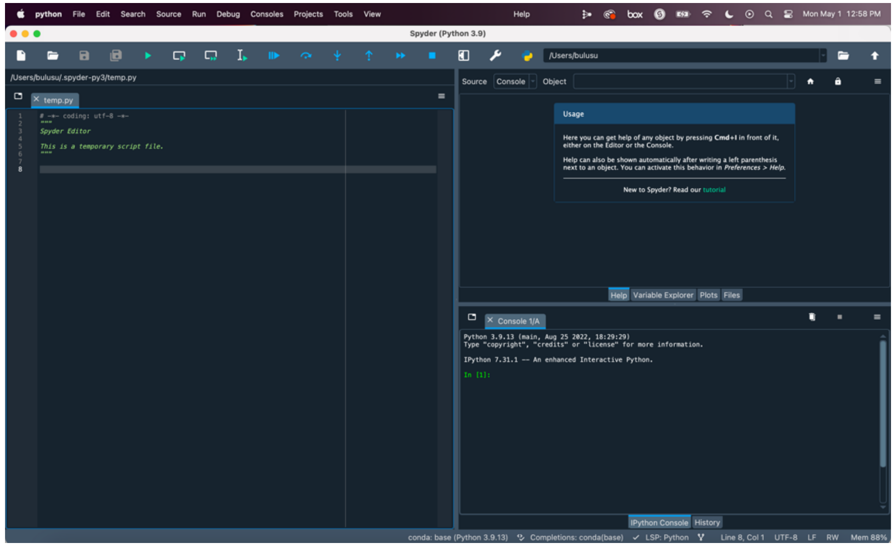
What are the other options?

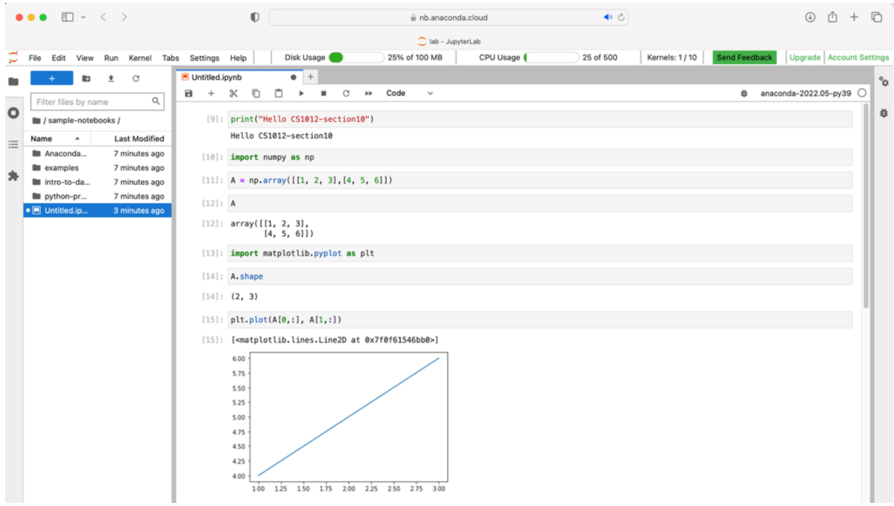
Source: Anaconda (IDE) https://www.anaconda.com

Anaconda python – IDEs:

- Spyder
- Jupyter Notebook
- Anaconda cloud

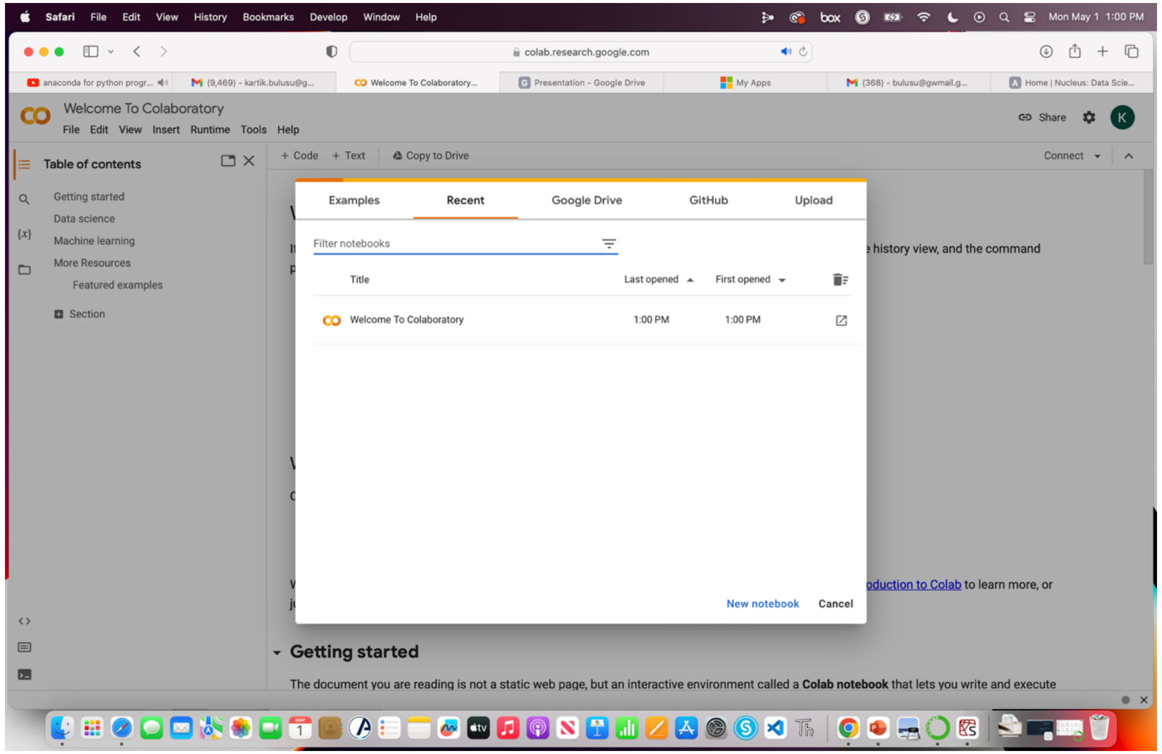






Google colab

Sources:
Google Colab https://colab.research.google.com/
Google Workspace Marketplace https://workspace.google.com/
Youtube video shorts https://youtube.com/shorts/7J3gU34j7Zk?feature=share



Colaboratory ("Colab" for short)

- Data analysis and machine learning tool that allows you to
  - combine executable Python code and
  - rich text along with charts, images, HTML, LaTeX and
  - more into a single document stored in Google Drive.

It connects to powerful Google Cloud Platform runtimes and enables you to easily share your work and collaborate with others.

Deepnote

Source: https://deepnote.com

Deepnote

Notebooks: a better way for teams to work with data

Jump right into Deepnote to explore data with Python and SQL, work together with your team, and share insights that lead to action — all from the comfort of your browser.

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More...

Sources:
Jetbrains-Pycharm: https://www.jetbrains.com/pycharm/
Enthought Deployment Manager (EDM): https://assets.enthought.com/downloads/